



UMS
UNIVERSITI MALAYSIA SABAH

KK34703 WEB ENGINEERING

GROUP PROJECT

(1ST SEMESTER 2023/2024)

BIKE RENTAL MANAGEMENT SYSTEM

PREPARED BY:

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1.0 Project Specification

#	Modules and submodules	Description	User Types	CRUD Operations	Priority
1	User Authentication 1.1 Registration 1.2 Login/Logout	This module manages user access to the system, allowing individuals to register, log in, and log out securely.	Admin, Customer	Create: Register new user records.	1
2	Bike Booking 2.1 Select Bicycle 2.2 Choose Booking Date and Duration 2.3 Confirm and Cancel Booking	This module facilitates the process of reserving bikes by providing users with options to choose available bikes, select one, choose a booking date and duration, and confirm the booking.	Customer	Create: Select a bike, choose booking date and duration, and confirm booking.	1
3	Booking History 3.1 View Past Bookings	This module is to keep track of users' past bike bookings, allowing them to view their booking history and their booking status.	Customer	Read: View past bookings.	1
4	Customer Information Management 4.1 Update Profile 4.2 View Personal Information	This module enables users to maintain and update their personal information, ensuring accurate and up-to-date details.	Admin, Customer	Update: Modify user profile information. Read: View personal information. Delete: Delete user records (restricted to Admin).	1
5	Feedback 5.1 Submit Feedback 5.2 View Feedbacks	This module allows users to submit feedback regarding their experiences.	Admin, Customer	Create: Submit feedback. Read: View feedback (restricted to	2

				Admin).	
6	Bookings report 6.1 View All Bookings 6.2 Filter Bookings by Booking Status 6.3 Update Booking Status	This module allows admin to view all the bike bookings. Admin can filter the bookings based on the booking status whether under processing, approved, rejected, in use or returned.	Admin	Read: View all bookings and filter bookings by booking status. Update: Modify booking status.	1
7	Bicycle Management 7.1 Add New Bicycles to Inventory 7.2 Remove Bicycles from Inventory 7.3 Update Bicycle Details 7.4 Search Bicycles by Name	This module is to manage the inventory of bikes by providing functions to add new bikes, remove bikes, update bike details and search bikes by name.	Admin	Create: Add new bikes to inventory. Delete: Remove bikes from inventory. Update: Modify bike details. Read: View and search bikes by name.	1
8	Payment 8.1 Make Payment 8.2 Payment Confirmation	This module handles payment transactions for bike bookings, allowing users to make payments and upload their receipt.	Customer	Create: Make a payment.	1

2.0 MySQL Database

2.1 Database Design

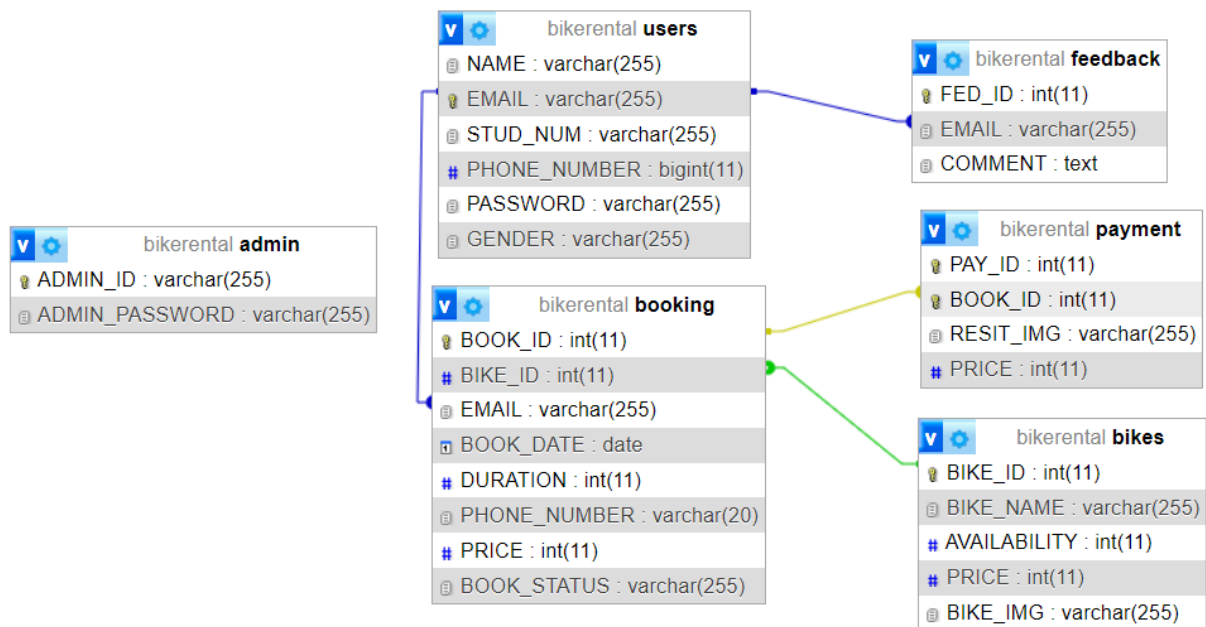


Table	Action	Rows	Type	Collation	Size	Overhead
☐ admin	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ bikes	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ booking	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ feedback	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ payment	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	27	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

2.2 Database Table, Column and Sample Data

2.2.1 Admin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 ADMIN_ID	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 ADMIN_PASSWORD	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More

ADMIN_ID

ADMIN_PASSWORD

☐ Edit Copy Delete ADMIN ADMIN

2.2.2 Users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 NAME	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 EMAIL	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 STUD_NUM	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 PHONE_NUMBER	bigint(11)			No	None			Change Drop More
☐	5 PASSWORD	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 GENDER	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More

NAME

EMAIL

STUD_NUM

PHONE_NUMBER

PASSWORD

GENDER

☐ Edit Copy Delete Danneal danneal@gmail.com BI21110018 60193521182 182511f9ada3effca3d8b3126d0fd0ff Male

☐ Edit Copy Delete Asnawie nawie@yahoo.com BI21110021 124789954 df697a98bf75ce70fd08dda39b8ad9b Male

☐ Edit Copy Delete Fairuz Ahmad pai@gmail.com BI21110017 60198313610 182511f9ada3effca3d8b3126d0fd0ff Female

2.2.3 Bikes

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 BIKE_ID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 BIKE_NAME	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 AVAILABILITY	int(11)			No	None			Change Drop More
☐	4 PRICE	int(11)			No	None			Change Drop More
☐	5 BIKE_IMG	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More

BIKE_ID

BIKE_NAME

AVAILABILITY

PRICE

BIKE_IMG

☐ Edit Copy Delete 1 Electric Bicycle 40 3 single_bike.jpg

☐ Edit Copy Delete 4 Electric Scooter 6 40 electric_scooter.jpg

☐ Edit Copy Delete 5 Electric Tandem Two Seats Bicycle 66 15 tandem.jpg

☐ Edit Copy Delete 33 Electric Mountain Bike 51 30 mountain_bike.jpg

2.2.4 Booking

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 BOOK_ID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 BIKE_ID	int(11)			No	None			Change Drop More
☐	3 EMAIL	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 BOOK_DATE	date			No	None			Change Drop More
☐	5 DURATION	int(11)			No	None			Change Drop More
☐	6 PHONE_NUMBER	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	7 PRICE	int(11)			No	None			Change Drop More
☐	8 BOOK_STATUS	varchar(255)	utf8mb4_general_ci		No	UNDER PROCESSING			Change Drop More

				BOOK_ID	BIKE_ID	EMAIL	BOOK_DATE	DURATION	PHONE_NUMBER	PRICE	BOOK_STATUS
☐	Edit Copy Delete			89	1	pai@gmail.com	2024-01-05	5	0198313610	3	RETURNED
☐	Edit Copy Delete			90	1	pai@gmail.com	2024-01-05	1	0198313610	3	IN USE
☐	Edit Copy Delete			92	1	pai@gmail.com	2024-01-09	2	0198313610	6	UNDER PROCESSING

2.2.5 Payment

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 PAY_ID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 BOOK_ID	int(11)			No	None			Change Drop More
☐	3 RESIT_IMG	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 PRICE	int(11)			No	None			Change Drop More

				PAY_ID	BOOK_ID	RESIT_IMG	PRICE
☐	Edit Copy Delete			37	89	BOOK_ID89.jpg	3
☐	Edit Copy Delete			38	90	BOOK_ID90.jpg	3
☐	Edit Copy Delete			39	92	BOOK_ID92.pdf	6

2.2.6 Feedback

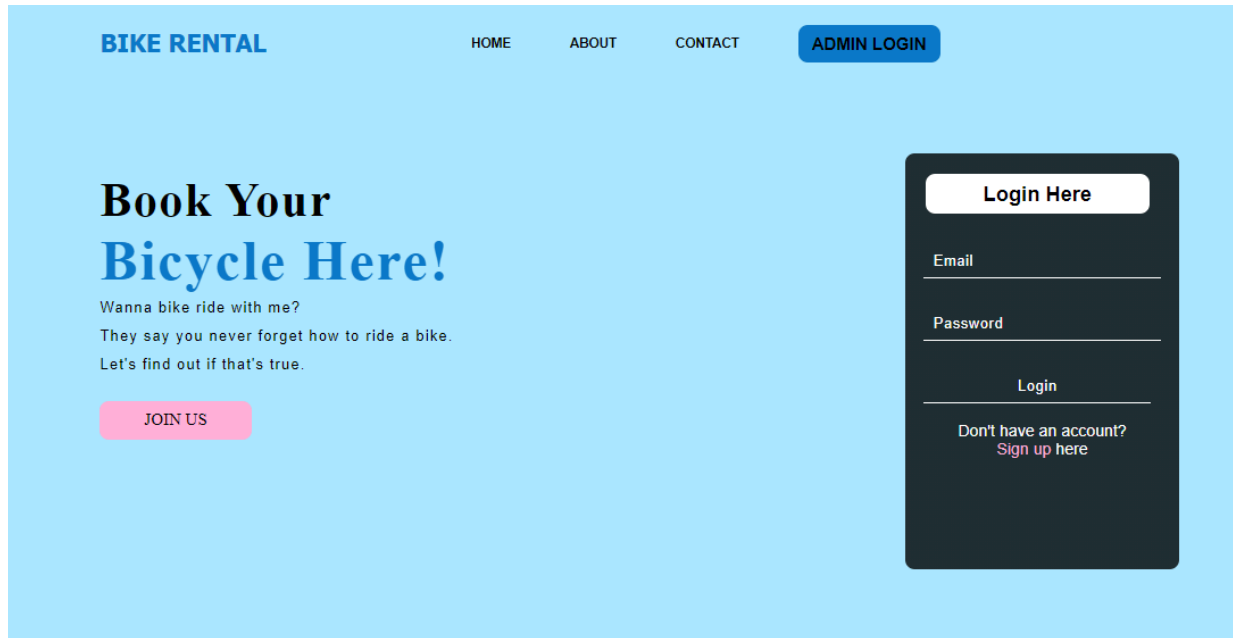
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 FED_ID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 EMAIL	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 COMMENT	text	utf8mb4_general_ci		No	None			Change Drop More

				FED_ID	EMAIL	COMMENT
☐	Edit Copy Delete			14	pai@gmail.com	bicycle tire have less air. please fix that. thank...
☐	Edit Copy Delete			15	danneal@gmail.com	Staff was very friendly.

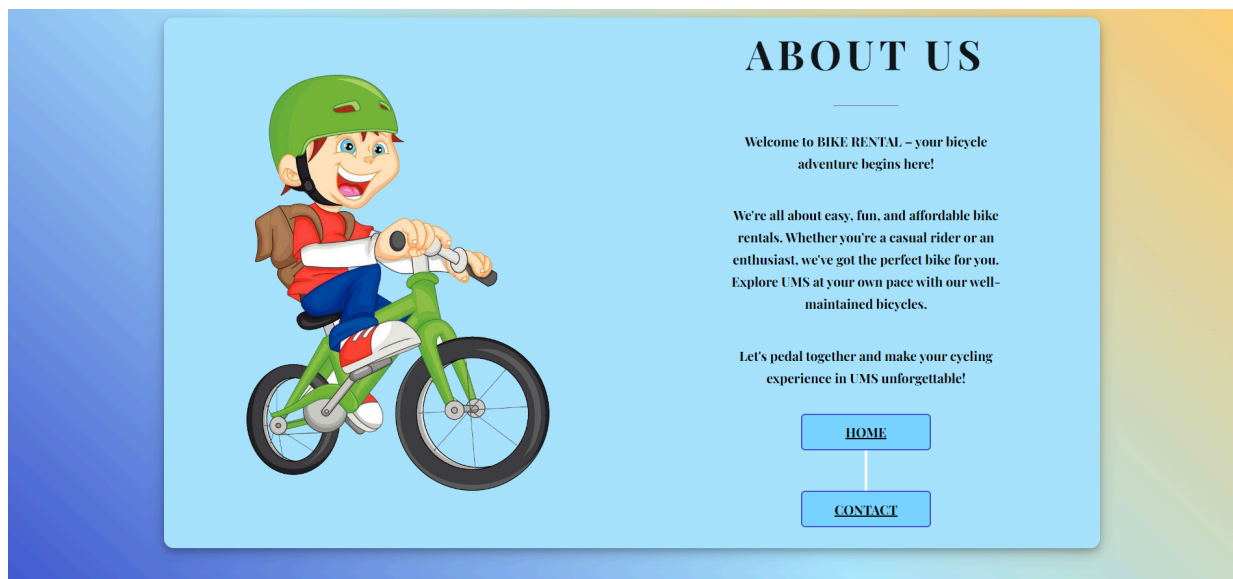
3.0 User Interface

3.1 User

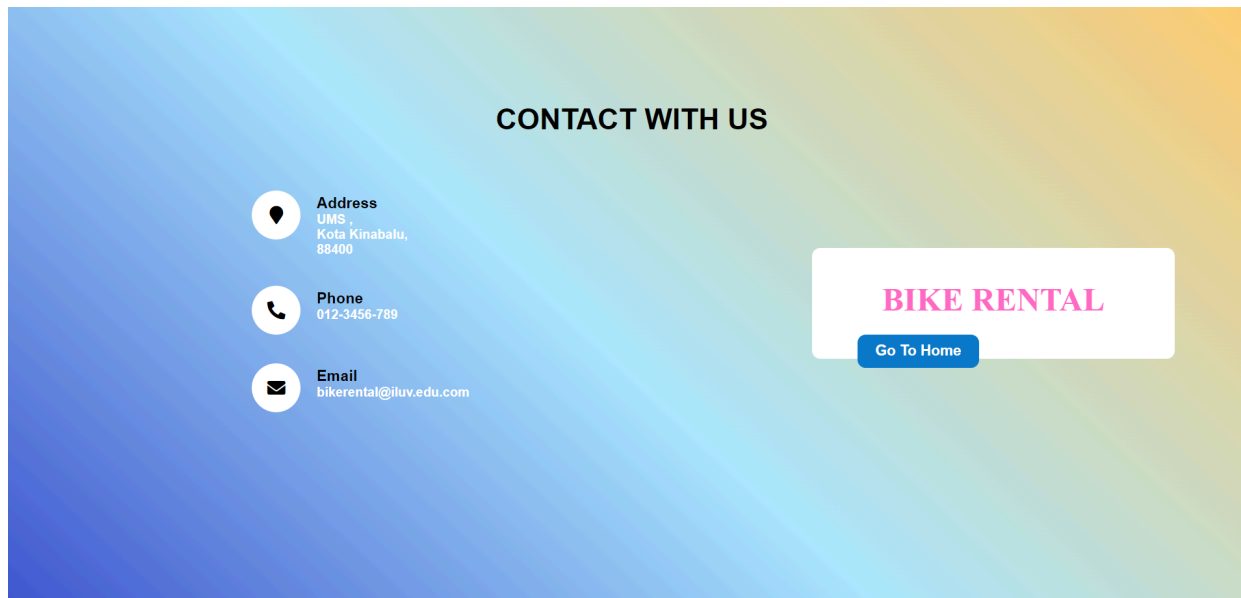
The main page will feature a login section for users.



When the user selects 'About Us'.



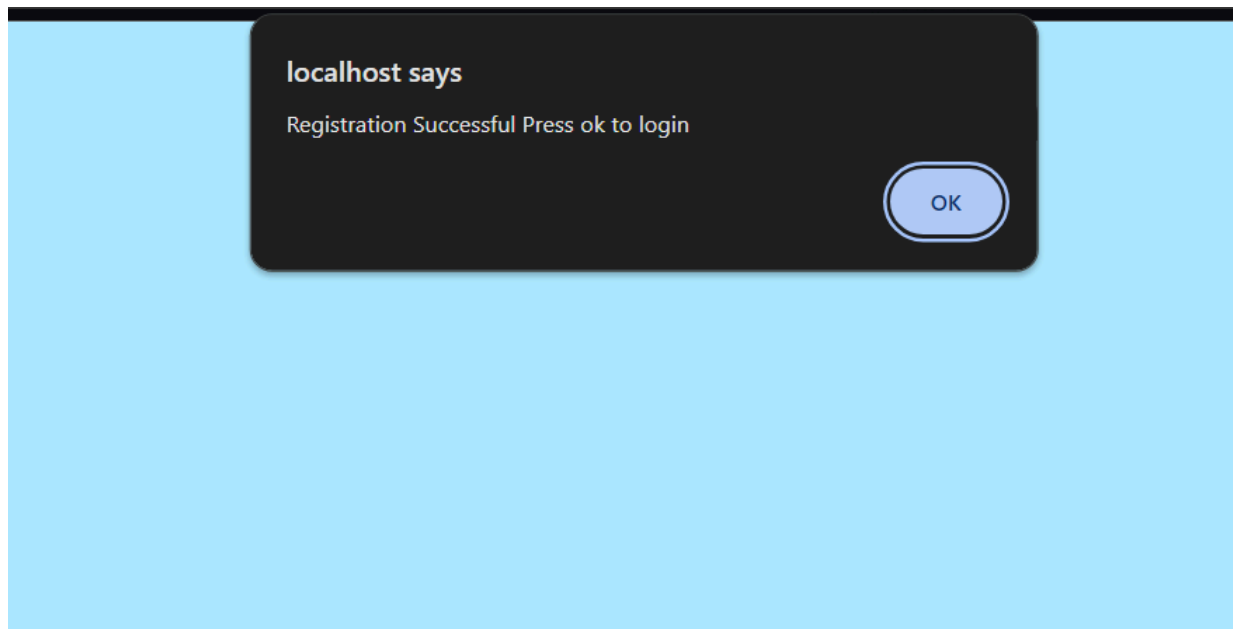
When the user selects 'Contact' they will be given access to the contact information for the bike rental shop.



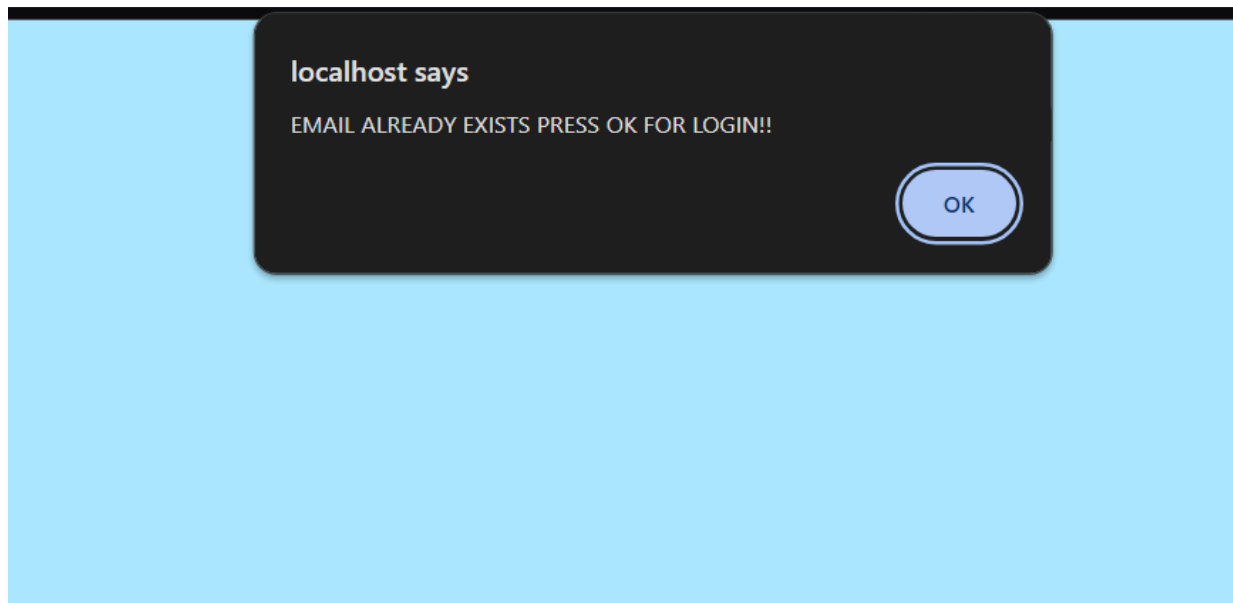
To register, users need to input their full name, email, student number, phone number, password, and choose a gender.

A screenshot of a registration form titled 'Register Here'. The form is set against a light blue background. At the top left, there is a pink button labeled 'HOME'. In the center, the text 'RENT YOUR BICYCLE NOW!' is displayed in pink. The registration form itself is a dark grey rectangle with white text and input fields. It contains the following fields: Full Name (with the example 'MUHAMMAD ASHAFIE BIN BUKAH'), Email (with the example 'ashafieashafie@gmail.com'), Student Number (with the example 'W21110021'), Phone Number (with the example '+60107602723'), Password (with a masked input), and Confirm Password (with a masked input). Below these fields is a 'Gender' section with radio buttons for 'Male' (selected) and 'Female'. At the bottom of the form is a pink button labeled 'REGISTER'.

This occurs upon successful user registration.



This occurs when the user tries to register with email that is already on record.



When a user is logged in, the website menu displays the available types of bikes that they can book.

BIKE RENTAL[HOME](#)[ABOUT](#)[CONTACT](#)[FEEDBACK](#)[LOGOUT](#)

HELLO!
MUHAMMAD
ASNAWIE BIN
BUKAH

[BOOKING
STATUS](#)

BICYCLE TYPES



Electric Bicycle
Availability : 40
Rent Per Hour : RM 3 /-
[BOOK](#)



Electric Scooter
Availability : 6
Rent Per Hour : RM 40 /-
[BOOK](#)

When users click on 'Feedback' they have the option to provide their feedback.

FEEDBACK

Name:
MUHAMMAD ASNAWIE BIN BUKAH

Email:
awieasnawie8@gmail.com

Message:
Bagus

[Submit Feedback](#)

[Home](#)

A confirmation message will be displayed to indicate that the feedback has been successfully submitted.

Feedback Submitted

Thank you for your feedback! We appreciate your comment.

[Back to Home](#)

This page is revealed when users click on their name in the top-right corner, displaying their profile information. From here, they have the option to edit their profile.

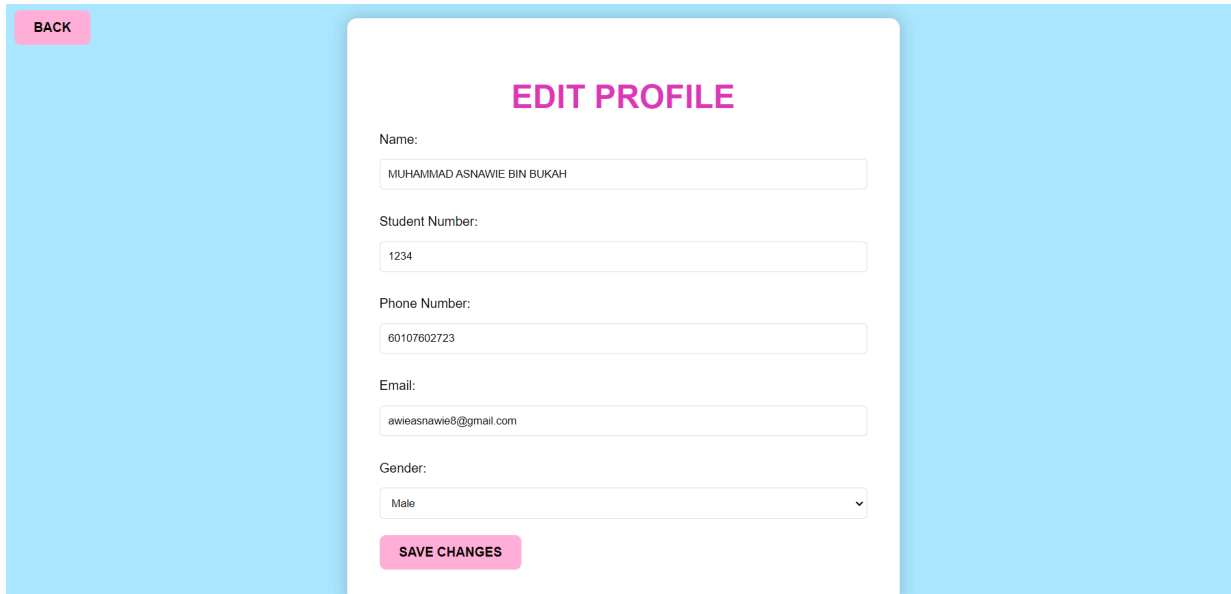
HOME

USER PROFILE

Username :	MUHAMMAD ASNAWIE BIN BUKAH
Email :	awieasnawie8@gmail.com
Student Number :	1234
Phone Number :	60107602723
Gender :	Male

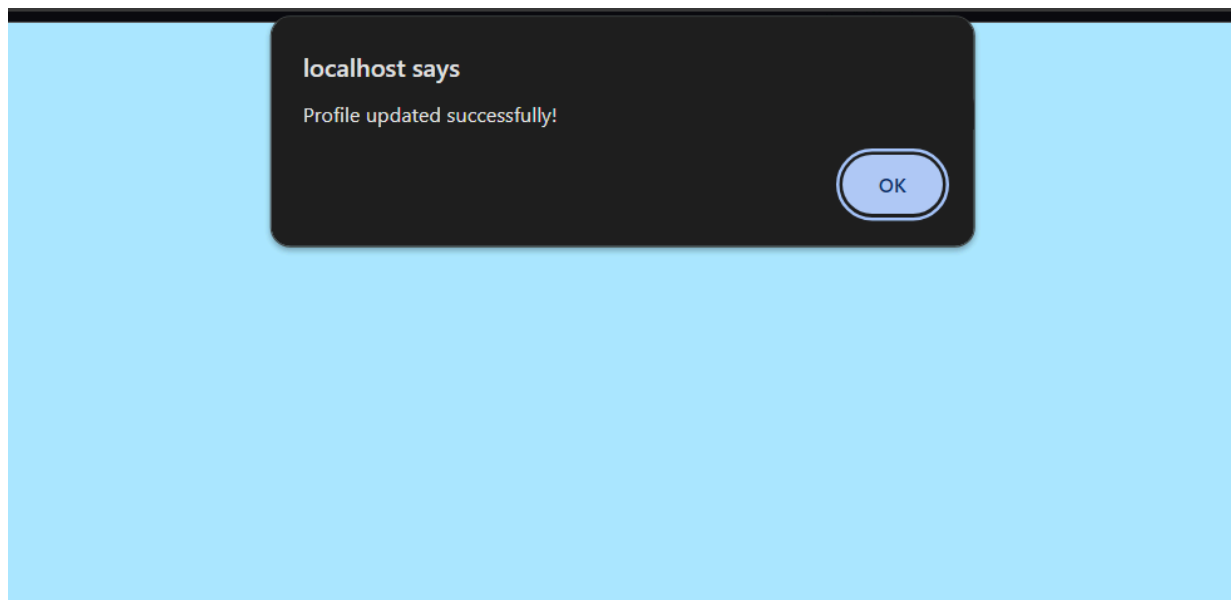
EDIT PROFILE

When the user clicks on 'Edit Profile' this page will be displayed.

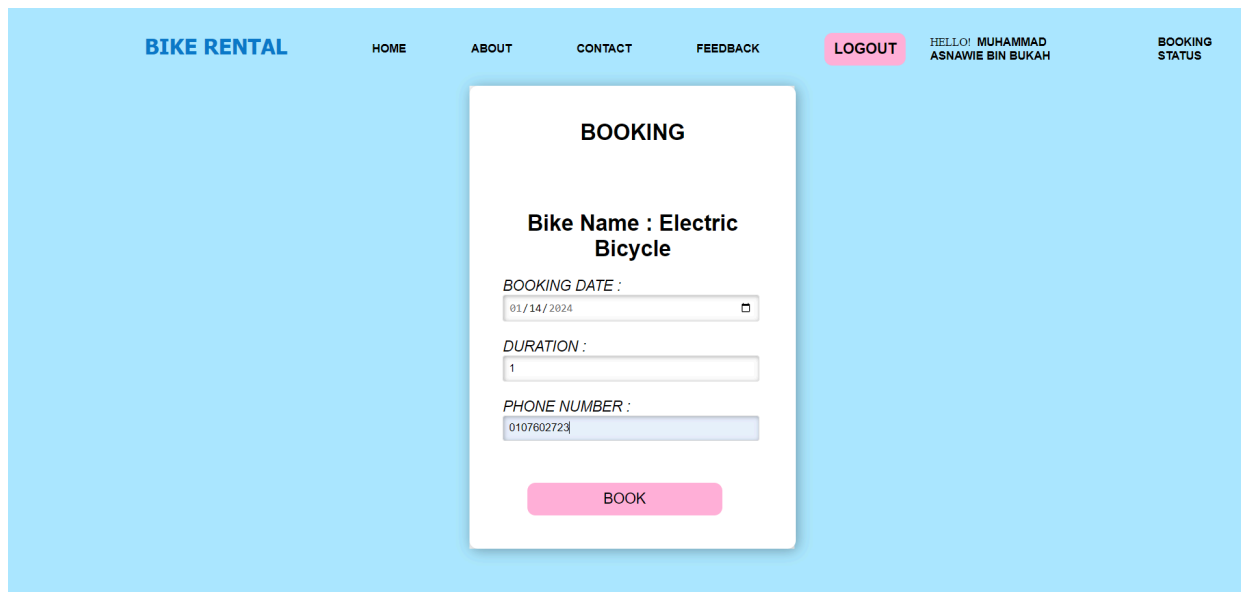


A screenshot of a web application showing an 'EDIT PROFILE' form. The form is centered on a light blue background. In the top left corner of the background, there is a pink button labeled 'BACK'. The form itself is a white card with a pink title 'EDIT PROFILE'. It contains several input fields: 'Name' with the value 'MUHAMMAD ASNAWIE BIN BUKAH', 'Student Number' with '1234', 'Phone Number' with '00107602723', and 'Email' with 'awieasnawie8@gmail.com'. There is also a 'Gender' dropdown menu currently set to 'Male'. At the bottom of the form is a pink button labeled 'SAVE CHANGES'.

This occurs when the profile has been successfully edited.

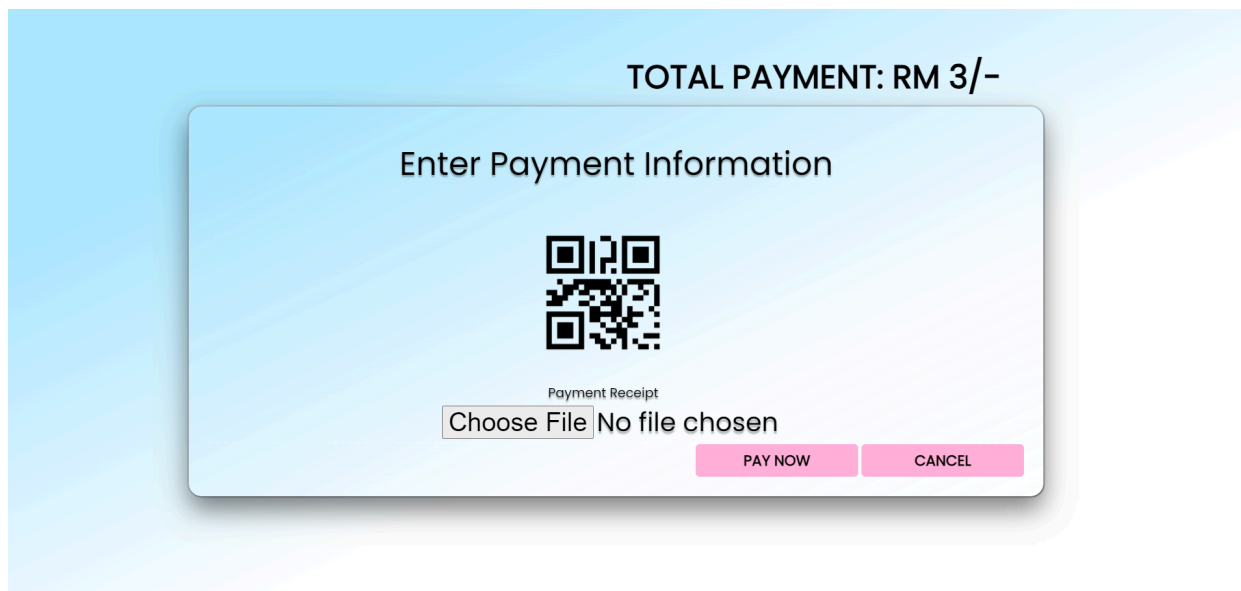


To make a booking, users need to click on 'Book' in the menu page and then fill out the required form.



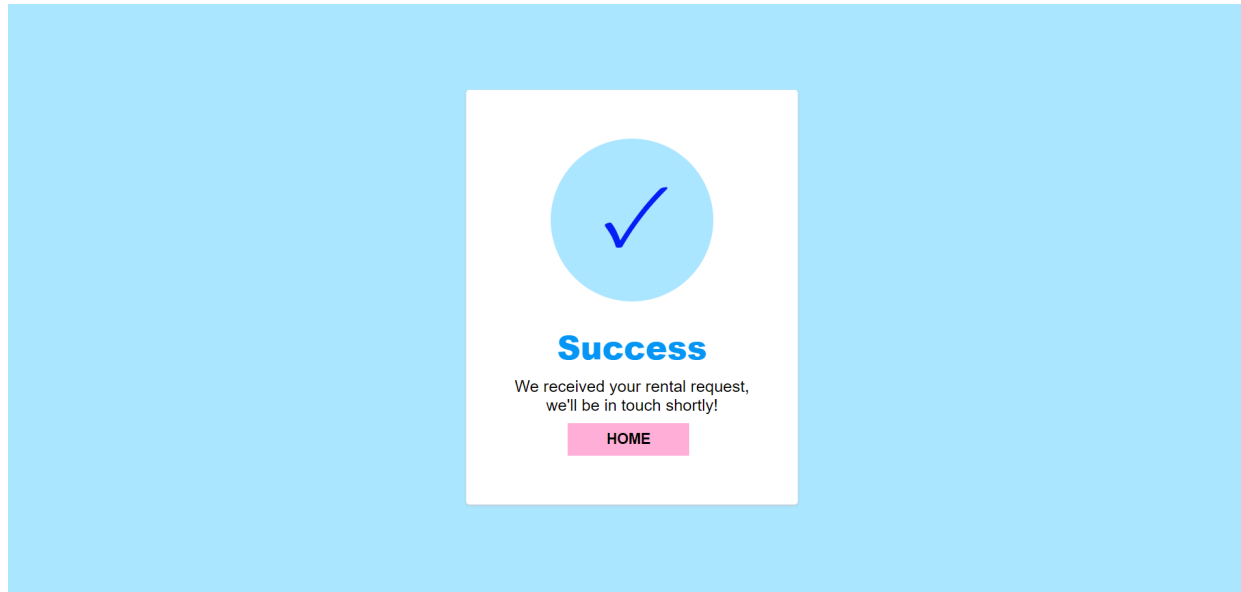
The screenshot shows a web application for 'BIKE RENTAL'. The header includes navigation links: HOME, ABOUT, CONTACT, FEEDBACK, and a pink LOGOUT button. On the right, it says 'HELLO! MUHAMMAD ASNAWIE BIN BUKAH' and 'BOOKING STATUS'. The main content is a white 'BOOKING' form. It displays 'Bike Name : Electric Bicycle'. Below this are three input fields: 'BOOKING DATE : 01/14/2024' with a calendar icon, 'DURATION : 1', and 'PHONE NUMBER : 0107602723'. At the bottom of the form is a pink 'BOOK' button.

Following that, users need to complete the payment process by scanning the QR code and making the payment. Subsequently, they are required to upload the payment receipt here.

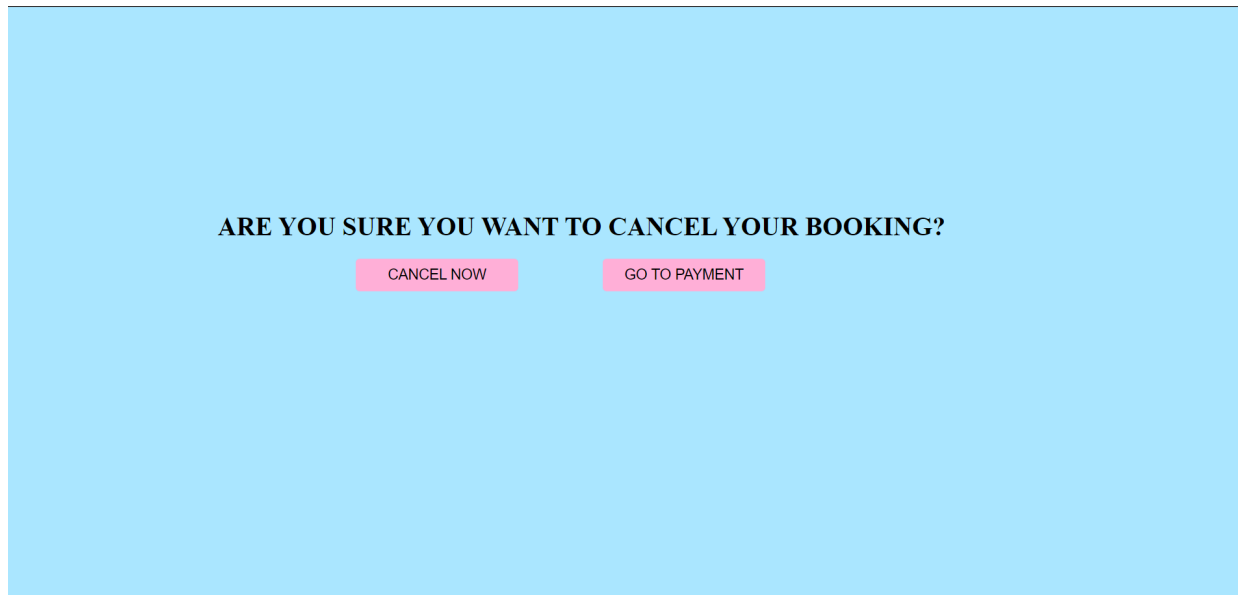


The screenshot shows a payment confirmation screen. At the top, it says 'TOTAL PAYMENT: RM 3/-'. Below this is a white box titled 'Enter Payment Information'. Inside the box, there is a QR code. Below the QR code, it says 'Payment Receipt' and 'Choose File No file chosen'. At the bottom of the box are two pink buttons: 'PAY NOW' and 'CANCEL'.

This indicates that the payment has been successfully processed.



If the user wishes to cancel their payment, the following page will appear. Upon cancellation, they will be redirected back to the main menu.



Users can view their booking status by clicking on 'Booking Status' located in the right corner of the main menu on the website page.

[Go to Home](#)

BIKE TYPE	DURATION	BOOK DATE	BOOKING STATUS
Electric Tandem Two Seats Bicycle	3 hour	2024-01-14	UNDER PROCESSING
Motor	3 hour	2024-01-14	UNDER PROCESSING
Electric Scooter	2 hour	2024-01-14	UNDER PROCESSING
Electric Tandem Two Seats Bicycle	1 hour	2024-01-14	APPROVED
Electric Scooter	1 hour	2024-01-14	APPROVED

To log out, the user can click the logout button, which will take them to the main home login page.

BIKE RENTAL

HOMEABOUTCONTACTFEEDBACK

LOGOUT

HELLO! MUHAMMAD ASNAWIE BIN BUKAH

BOOKING STATUS

BICYCLE TYPES

BIKE RENTAL

HOMEABOUTCONTACT

ADMIN LOGIN

Book Your Bicycle Here!

Wanna bike ride with me?
They say you never forget how to ride a bike.
Let's find out if that's true.

JOIN US

Login Here

Email

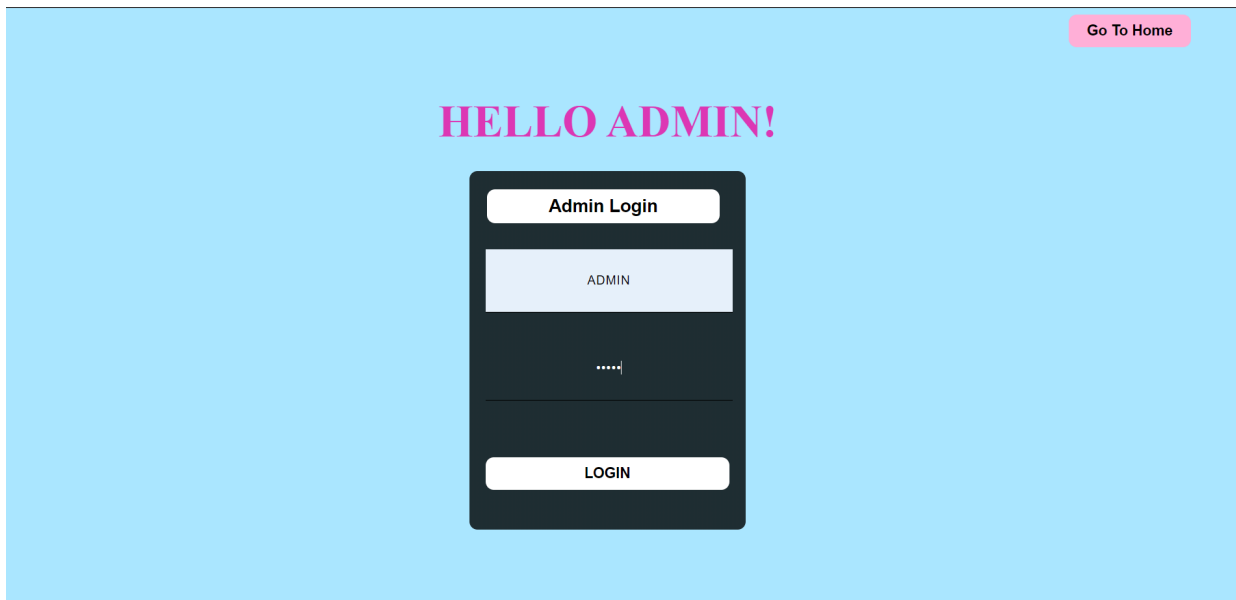
Password

Login

Don't have an account?
Sign up here

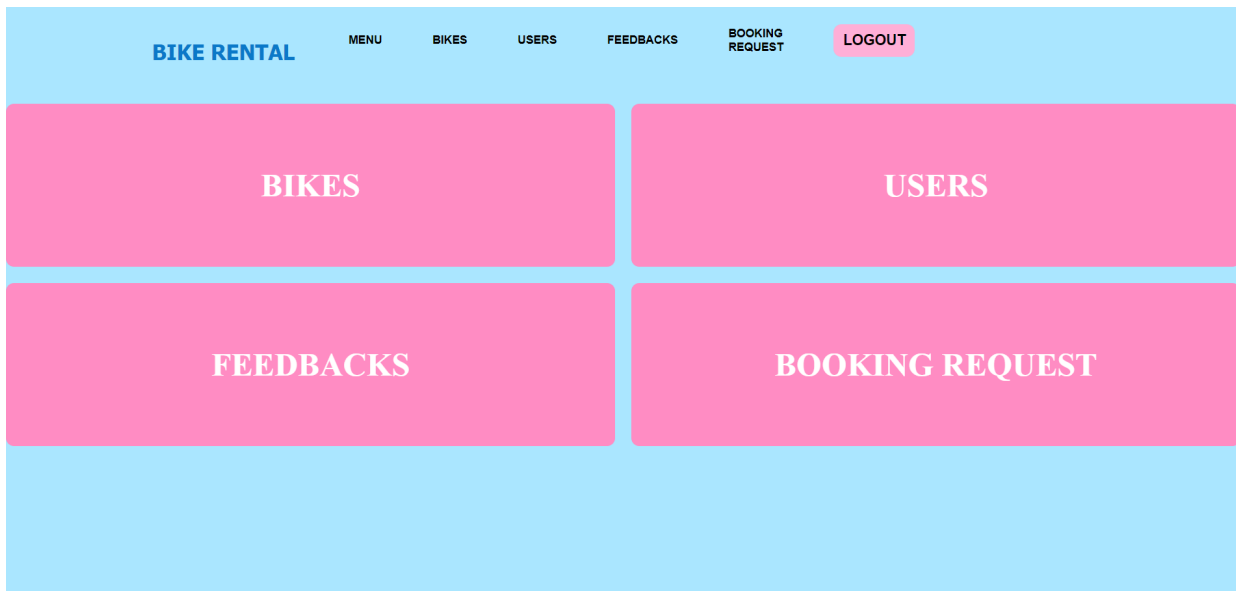
3.2 Admin

This is the login page designed for administrators.



The image shows a login page for administrators. It features a light blue background. In the top right corner, there is a pink button labeled "Go To Home". In the center, there is a dark grey rectangular box containing the login form. The form has a white header with the text "Admin Login". Below this, there is a light blue input field labeled "ADMIN". Underneath the input field is a dark grey field for a password, represented by dots. At the bottom of the form is a white button labeled "LOGIN". Above the form, the text "HELLO ADMIN!" is displayed in a bold, pink, serif font.

This is the main menu of the admin page.



This is the bike management page for administrators. Admins have the ability to delete, update, or add new types of bikes here.

The screenshot shows the 'BIKE MANAGEMENT' page. At the top, there is a navigation bar with 'BIKE RENTAL' and links for 'MENU', 'BIKES', 'USERS', 'FEEDBACKS', 'BOOKING REQUEST', and 'LOGOUT'. Below the navigation bar, the page title 'BIKE MANAGEMENT' is centered. To the right of the title is a '+ ADD BIKE' button. Below the title, there is a search bar labeled 'Search Bike:' with a placeholder 'Enter Bike Name', a 'Search' button, and a 'Reset' button. Below the search bar is a table with the following columns: 'NO', 'BIKE NAME', 'PRICE', 'AVAILABILITY', 'UPDATE', and 'DELETE'. The table contains four rows of bike data:

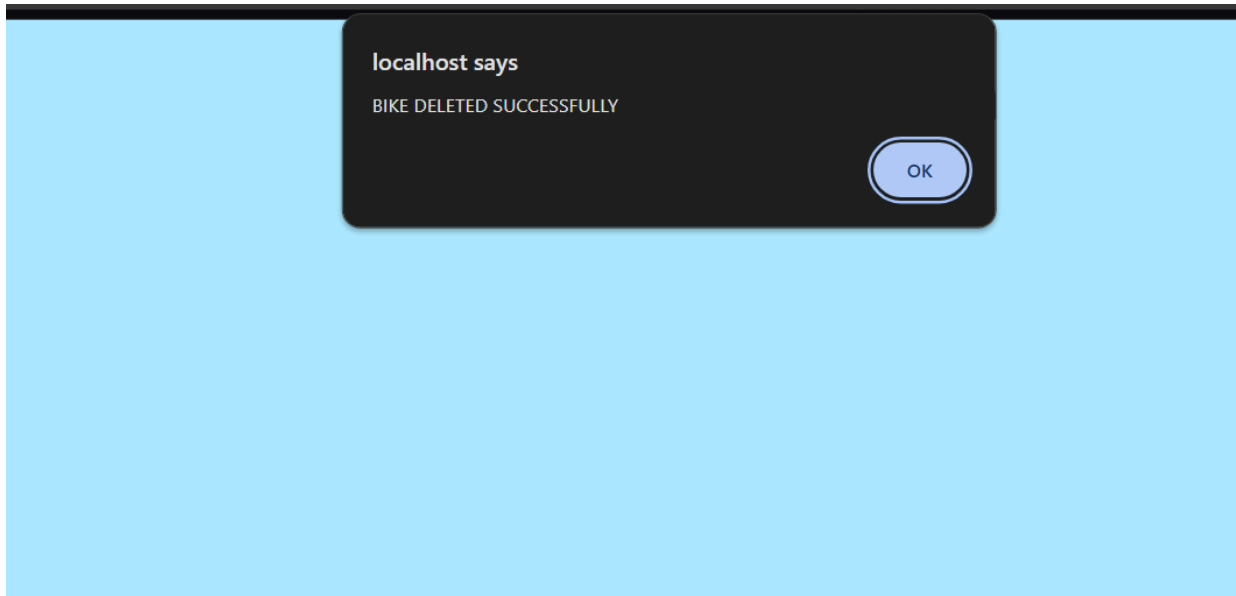
NO	BIKE NAME	PRICE	AVAILABILITY	UPDATE	DELETE
1	Electric Bicycle	RM 3	40	<button>UPDATE</button>	<button>DELETE</button>
2	Electric Scooter	RM 40	6	<button>UPDATE</button>	<button>DELETE</button>
3	Electric Tandem Two Seats Bicycle	RM 15	66	<button>UPDATE</button>	<button>DELETE</button>
4	Electric Mountain Bike	RM 30	51	<button>UPDATE</button>	<button>DELETE</button>

Admins can use the search feature by entering the bike name to quickly locate bikes for efficient searching.

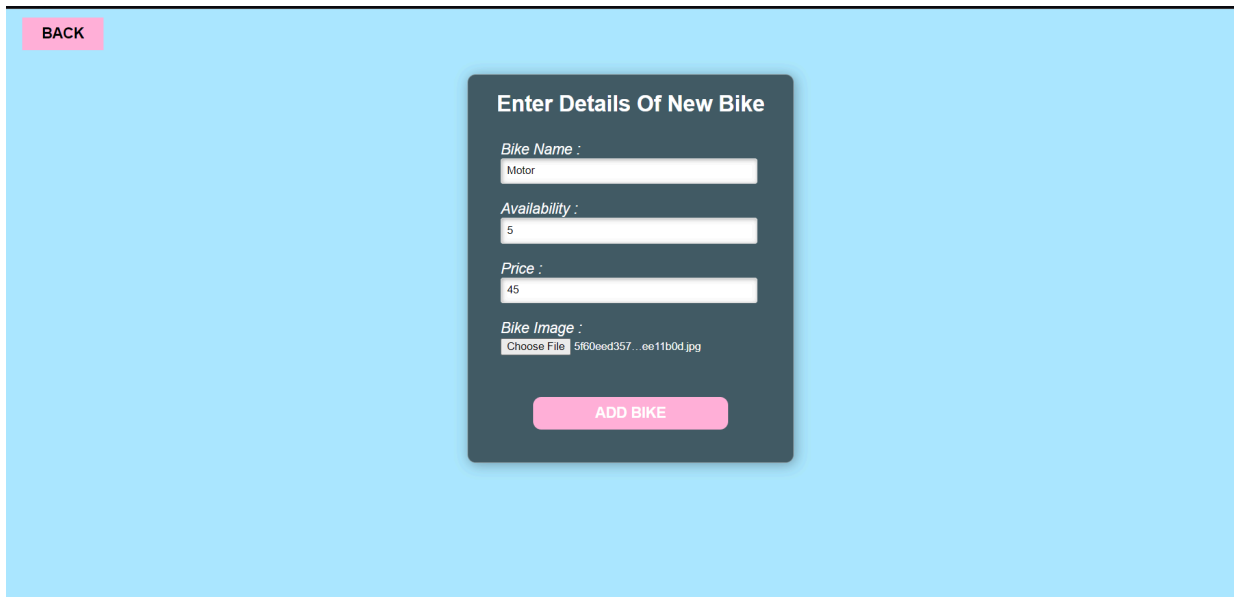
The screenshot shows the 'BIKE MANAGEMENT' page with the search bar filled with 'Two'. The 'Search' button is highlighted, and the table below shows the search results. The table has the same columns as the previous screenshot: 'NO', 'BIKE NAME', 'PRICE', 'AVAILABILITY', 'UPDATE', and 'DELETE'. The table contains one row of bike data:

NO	BIKE NAME	PRICE	AVAILABILITY	UPDATE	DELETE
1	Electric Tandem Two Seats Bicycle	RM 15	66	<button>UPDATE</button>	<button>DELETE</button>

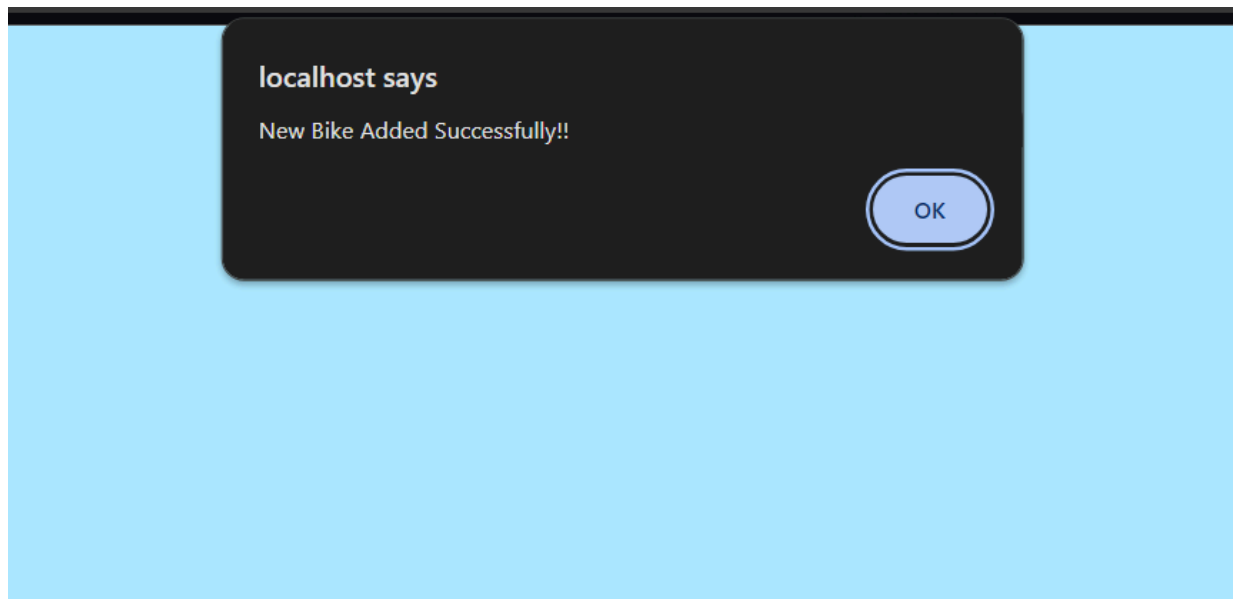
This occurs when the bike has been successfully deleted.



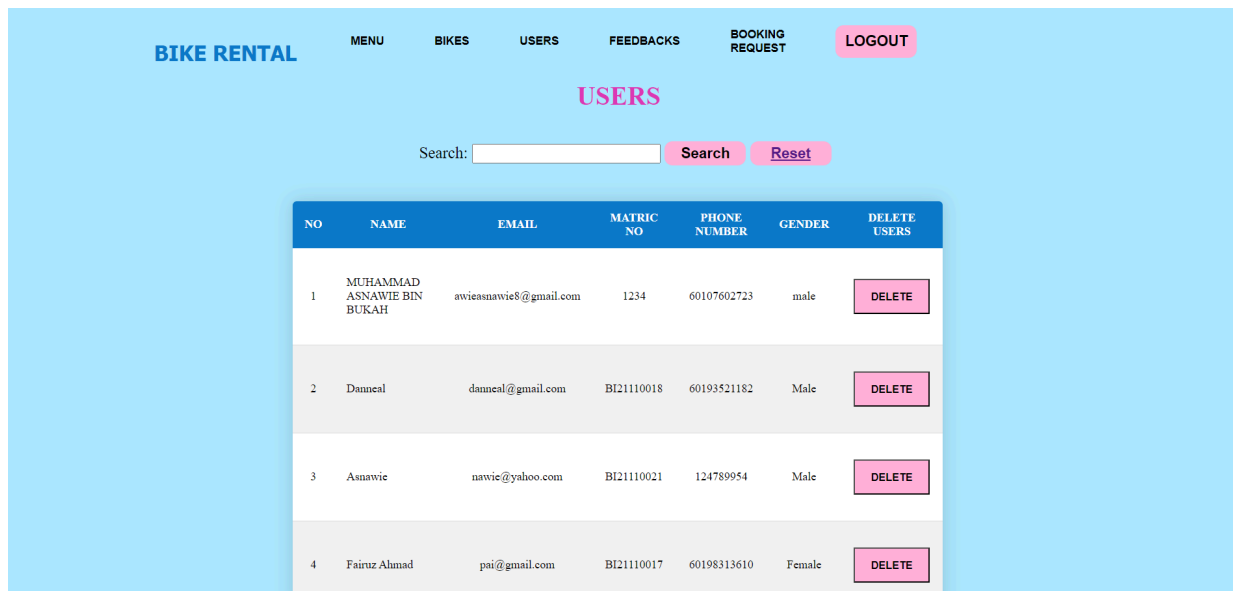
To add a new bike, administrators need to input the bike name, availability, price, and upload an image for the bike.

The image shows a web interface for adding a new bike. In the top left corner, there is a pink button labeled "BACK". The main content area is a light blue rectangle. Centered within this area is a dark gray form titled "Enter Details Of New Bike". The form contains four input fields: "Bike Name" with the value "Motor", "Availability" with the value "5", "Price" with the value "45", and "Bike Image" which includes a "Choose File" button and a filename "5f60eed357...ee11b0d.jpg". At the bottom of the form is a pink button labeled "ADD BIKE".

This confirms that the new bike has been successfully added.



In the Users page on the admin page, administrators have the capability to search and delete users.



This occurs when the user profile has been successfully deleted.

localhost says

USER DELETED SUCCESSFULLY

OK

On the feedback page, administrators have the ability to review customer feedback.

BIKE RENTAL

MENU

BIKES

USERS

FEEDBACKS

BOOKING
REQUEST

LOGOUT

FEEDBACKS

NO	EMAIL	COMMENT
1	danneal@gmail.com	Staff was very friendly.
2	pai@gmail.com	bicycle tire have less air. please fix that. thanks

On the booking request page within the admin site, administrators can update the booking status to approve, rejected, in use or returned.

BIKE RENTAL

MENUMENUBIKESUSERSFEEDBACKSBOOKING REQUESTLOGOUT

BOOKINGS

Search: -- Select Status -- Search Reset

NO	EMAIL	PHONE NUMBER	BOOK DATE	DURATION	PRICE	RECEIPT	BOOKING STATUS	APPROVE	REJECTED	IN USE	RETURNED
1	awieasnawie8@gmail.com	0107602723	2024-01-14	1 hour	RM 15	BOOK_ID99.PNG	APPROVED	APPROVE	REJECTED	IN USE	RETURNED
2	awieasnawie8@gmail.com	0107602723	2024-01-14	1 hour	RM 40	BOOK_ID28.jpg	UNDER PROCESSING	APPROVE	REJECTED	IN USE	RETURNED

Administrators have the ability to search for specific bookings by filtering their booking status.

BIKE RENTAL

MENUMENUBIKESUSERSFEEDBACKSBOOKING REQUESTLOGOUT

BOOKINGS

Search: -- Select Status -- Search Reset

-- Select Status --

UNDER PROCESSING

APPROVED

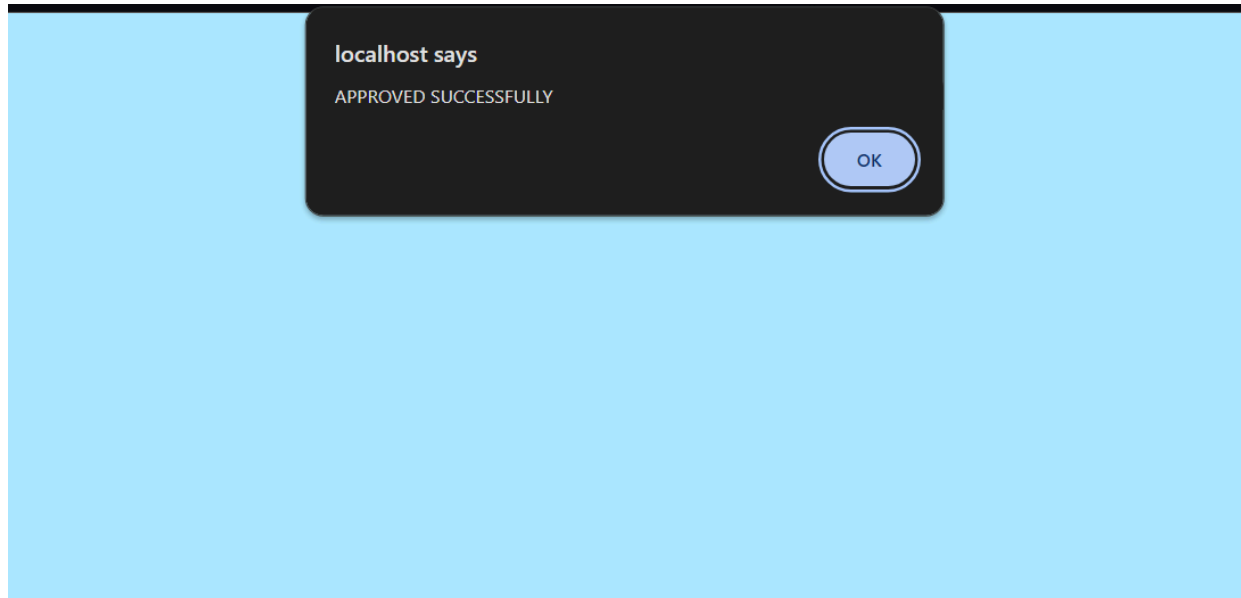
REJECTED

IN USE

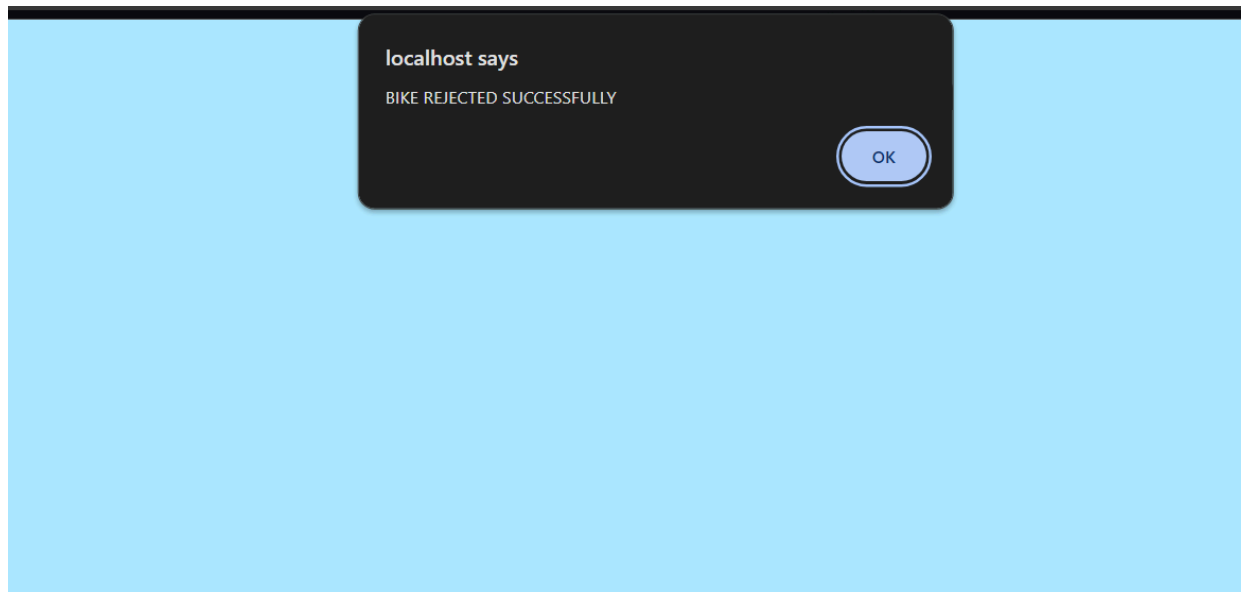
RETURNED

NO	EMAIL	PHONE NUMBER	BOOK DATE	DURATION	PRICE	RECEIPT	BOOKING STATUS	APPROVE	REJECTED	IN USE	RETURNED
1	awieasnawie8@gmail.com	0107602723	2024-01-14	1 hour	RM 15	BOOK_ID99.PNG	APPROVED	APPROVE	REJECTED	IN USE	RETURNED
2	awieasnawie8@gmail.com	0107602723	2024-01-14	1 hour	RM 40	BOOK_ID28.jpg	UNDER PROCESSING	APPROVE	REJECTED	IN USE	RETURNED

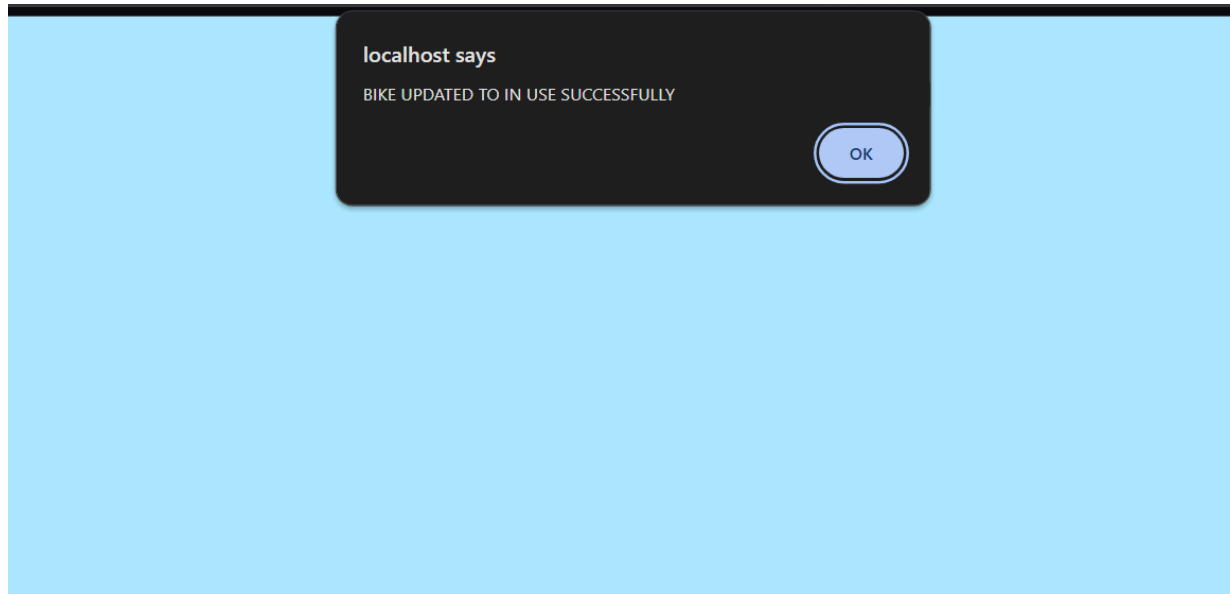
This occurs when the admin has approved the request.



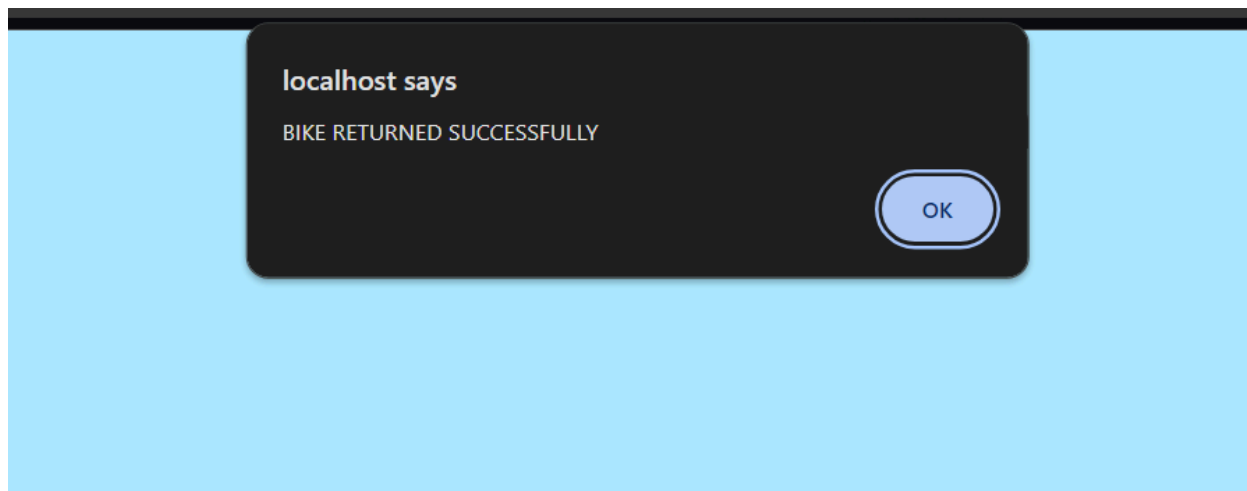
This occurs when the admin has rejected the request.



This occurs when the status for the booking is updated to in use.



This occurs when the admin has marked the bike as returned.



To log out, the admin can click the logout button, which will take them to the home main page.

